

Olist Business Discovery

HVIA Data & AI Solutions Internship - Task 01

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Purpose: understand Olist, analyze the public e-commerce dataset, identify business frictions, and propose a practical HVIA solution.

1. Executive Summary

- Olist is a Brazilian retail technology company that helps sellers operate online through a connected ecosystem of marketplaces, e-commerce tools, logistics, payments, and business services.
- The dataset contains 99441 orders from 2016-09-04 to 2018-10-17; 96478 orders were delivered.
- The strongest business pattern found is the connection between delivery performance and customer satisfaction.
- Late delivered orders had an average review score of 2.57, compared with 4.29 for on-time/early delivered orders.
- The proposed HVIA opportunity is a Logistics & Customer Satisfaction Intelligence Dashboard that connects delivery timing, seller performance, geography, categories, freight burden, and reviews.

2. Company Understanding

Olist operates as a retail technology ecosystem for sellers. The core business idea is to help small and medium sellers sell online and manage operations without needing to build all systems by themselves.

From a business discovery perspective, the key Olist stakeholders are sellers, customers, logistics partners, marketplaces, and internal operations teams. Seller success depends on marketplace access, smooth order management, reliable delivery, payment processing, and customer satisfaction.

This makes the dataset useful because it includes orders, customers, sellers, products, payments, freight, delivery timestamps, and reviews. These areas are directly connected to seller growth and customer experience.

3. Dataset Understanding

A key technical lesson in this dataset is row level. Not every table represents the same thing. Some tables are one row per order, while others can have multiple rows per order. This affects how tables should be joined.

Table	One row represents	Important keys
orders	One order with status and timeline	order_id, customer_id
customers	Customer record linked to an order	customer_id, customer_unique_id
order_items	One item inside an order	order_id, product_id, seller_id
products	One product and its category/details	product_id
sellers	One seller and seller location	seller_id
payments	One payment record for an order	order_id
reviews	One review record for an order	order_id
category_translation	Portuguese category to English category	product_category_name

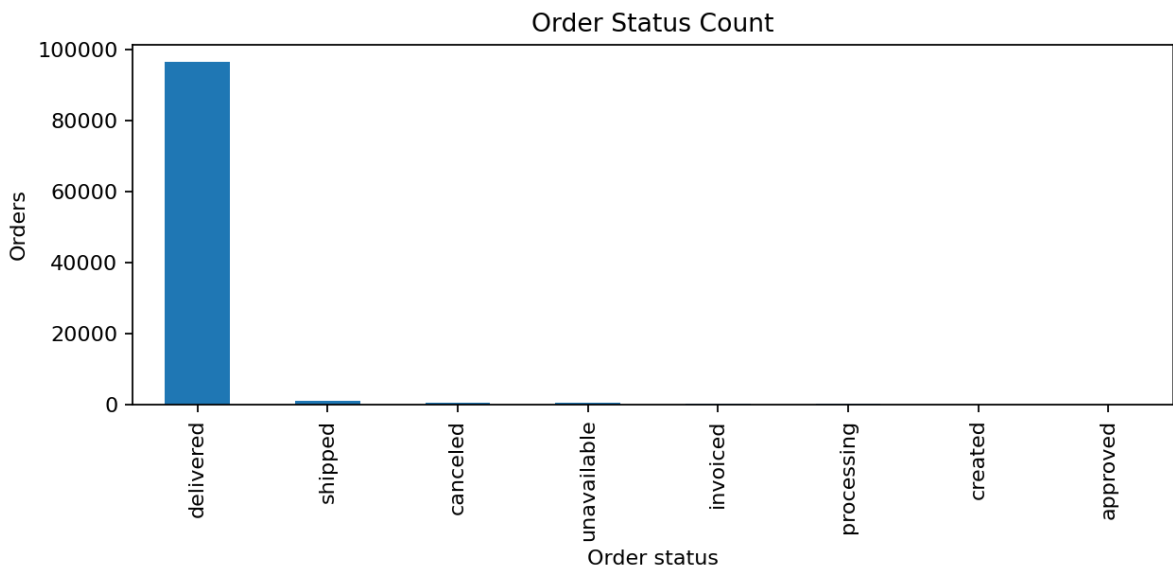
Important join warning: order_items and payments can contain multiple rows per order. Joining them directly to orders can duplicate payment or freight values. To avoid this, item, payment, and review data were aggregated to order level before joining.

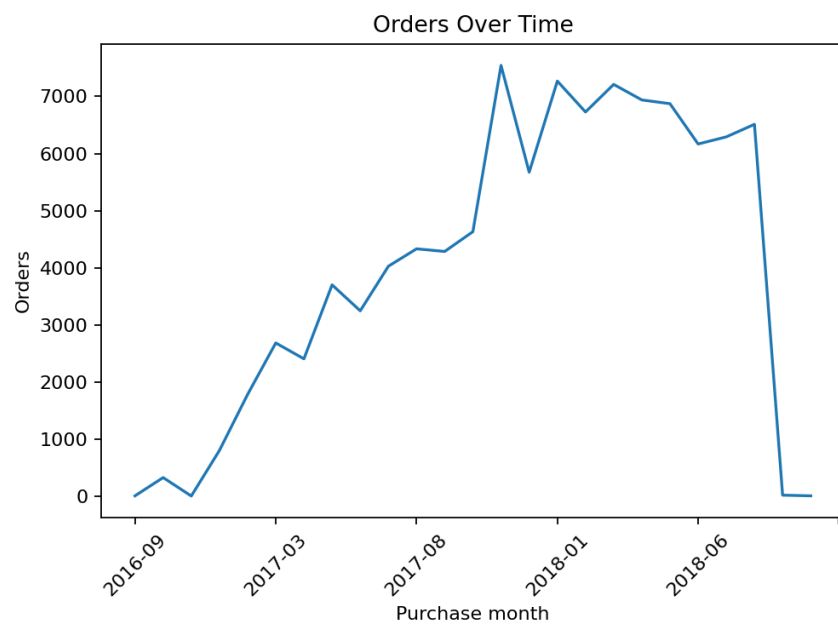
4. Technical Method

Step	What was done	Why it matters
1	Loaded all CSV files using pandas	Created a reproducible workflow
2	Checked row counts and unique IDs	Identified row level and repeated IDs
3	Converted timestamp columns to datetime	Allowed delivery and delay calculations
4	Merged product categories and seller locations onto item rows	Enabled category and seller analysis
5	Aggregated item/payment/review tables to order level	Avoided duplicated metrics
6	Built one order-level analysis table	Made business metrics reliable
7	Calculated delivery days, delay days, late flag, freight ratio	Created business-ready metrics

5. Business Overview

Metric	Value
Total orders	99441
Delivered orders	96478
Canceled orders	625
Unavailable orders	609
Total payment value	16,008,872.12
Average payment value	160.99
Unique customers	96096
Unique sellers	3095
Average review score	4.09
Average delivery days (delivered orders)	12.56
Late delivery rate (delivered orders)	8.11%
Repeat customer rate	3.12%



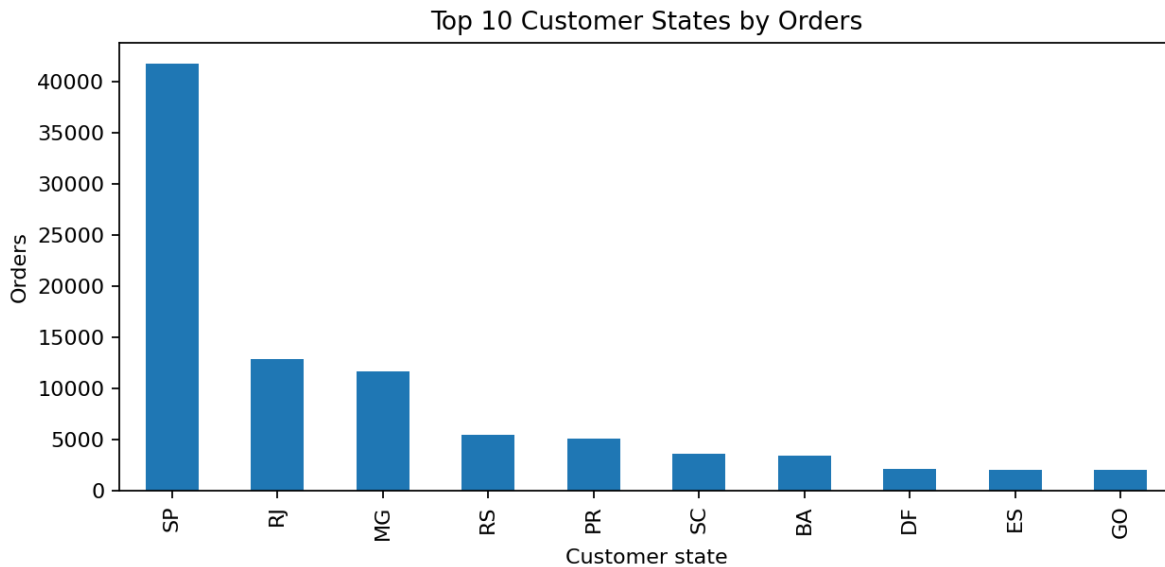


6. Location and Category Patterns

Customer location concentration

Order demand is highly concentrated in a few Brazilian states. SP is the largest customer state by order volume, followed by RJ and MG. This matters because operational improvements in high-volume states may affect a large share of orders.

Customer state	Orders
SP	41746
RJ	12852
MG	11635
RS	5466
PR	5045
SC	3637
BA	3380
DF	2140
ES	2033
GO	2020

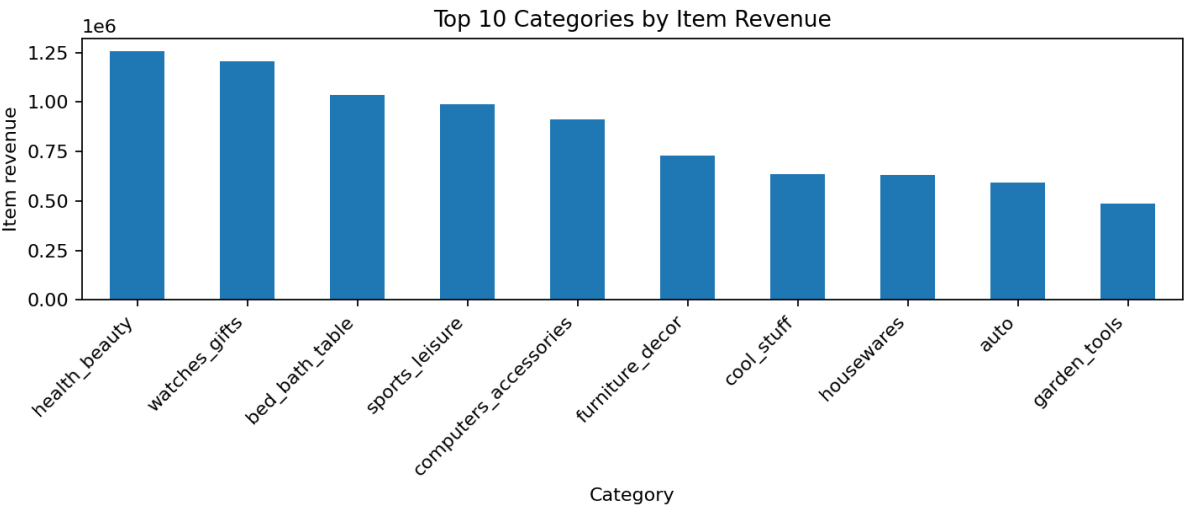


Category performance

Category analysis shows which product areas drive item revenue. This is useful because operational issues may not affect every category equally; some categories may have higher freight burden or stronger revenue importance.

Category	Unique orders	Item revenue	Freight ratio
health_beauty	8836	1,258,681.34	0.15
watches_gifts	5624	1,205,005.68	0.08
bed_bath_table	9417	1,036,988.68	0.20
sports_leisure	7720	988,048.97	0.17
computers_accessories	6689	911,954.32	0.16
furniture_decor	6449	729,762.49	0.24
cool_stuff	3632	635,290.85	0.13

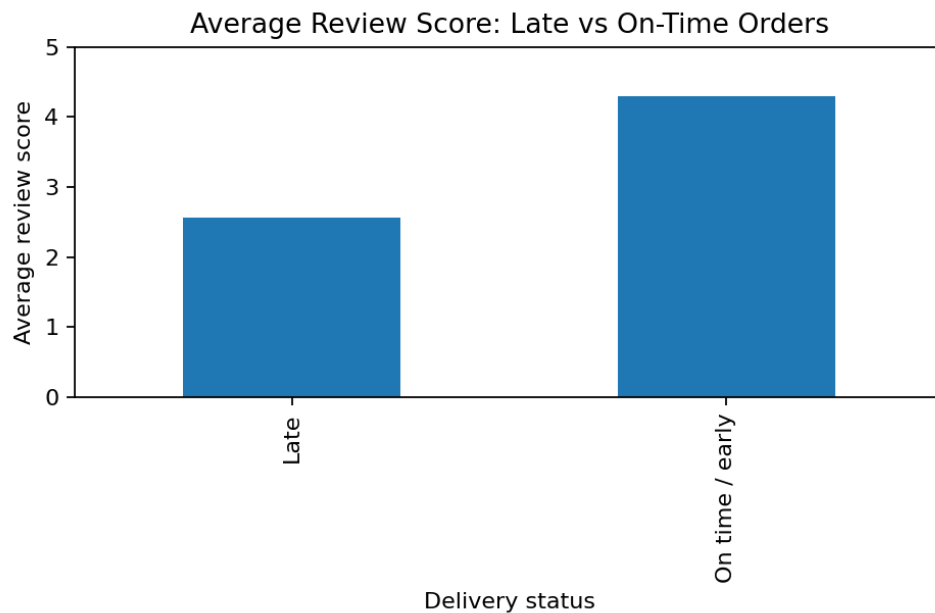
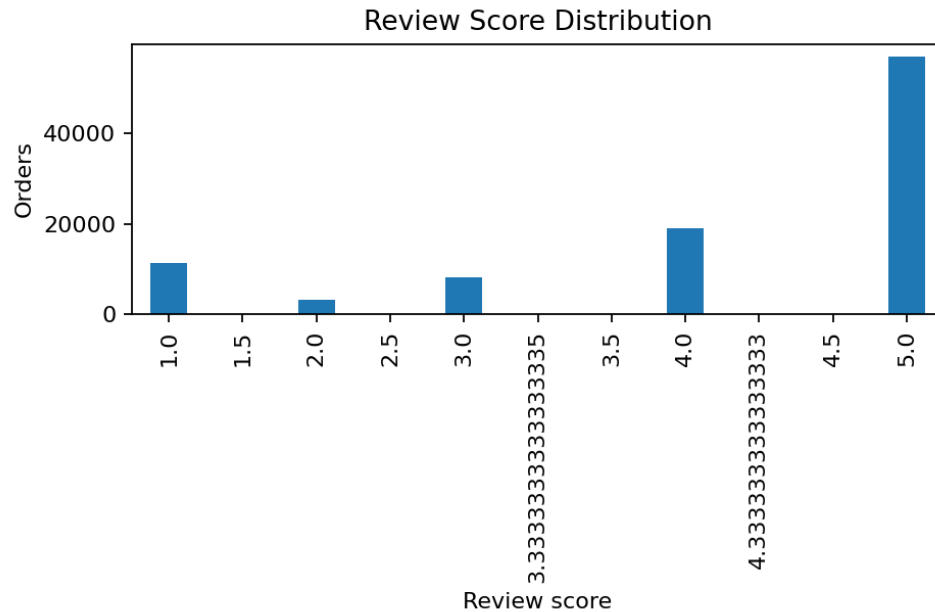
housewares	5884	632,248.66	0.23
auto	3897	592,720.11	0.16
garden_tools	3518	485,256.46	0.20



7. Main Finding: Delivery Performance and Reviews

The strongest finding is that late delivered orders have much lower review scores than on-time or early delivered orders. This does not prove delivery delay is the only cause of bad reviews, but it is a strong operational signal worth monitoring.

Delivery status	Orders	Avg review score	Avg delivery days	Avg delay days
Late	7,826	2.57	31.52	9.55
On time / early	88,652	4.29	10.88	-13.01

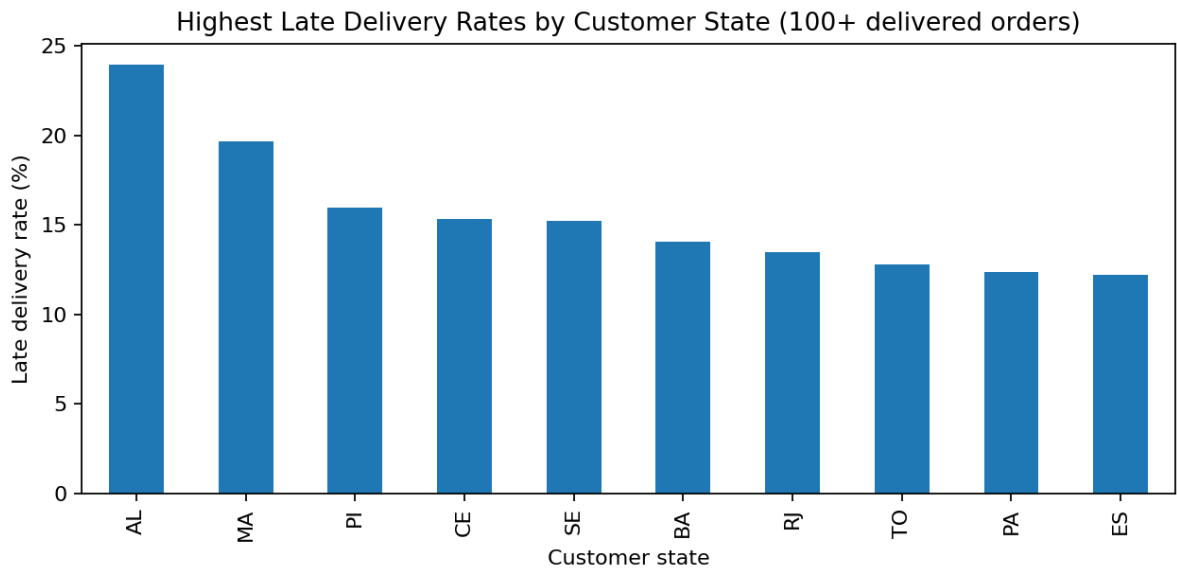


Late delivery by state

Late delivery rates also vary by customer state. This suggests a possible logistics intelligence use case: identify states, sellers, or routes where delivery risk is higher and customer satisfaction may be affected.

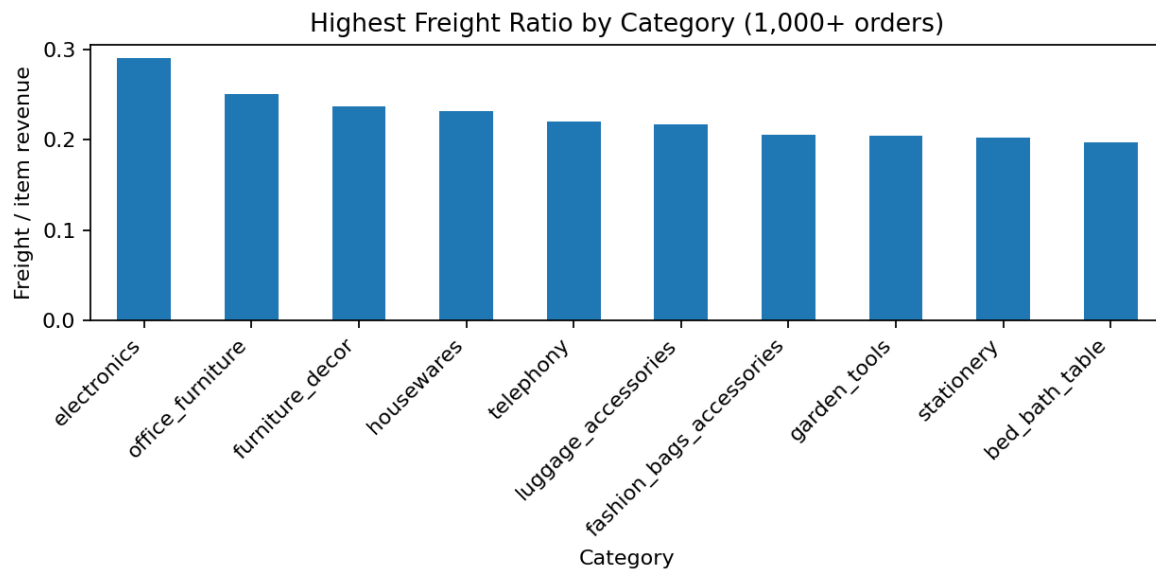
State	Orders	Avg delivery days	Late rate %	Avg review
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AL	397	24.54	23.93	3.85
MA	717	21.57	19.67	3.83
PI	476	19.46	15.97	3.99
CE	1,279	21.27	15.32	3.94
SE	335	21.52	15.22	3.91
BA	3,256	19.34	14.04	3.93
RJ	12,350	15.31	13.47	3.97
TO	274	17.66	12.77	4.15
PA	946	23.77	12.37	3.91
ES	1,995	15.79	12.23	4.08



8. Freight Burden

Freight burden was measured as freight value divided by item revenue. This is not a full profitability metric because internal cost/margin data is missing, but it is useful as a friction signal. High freight ratio categories may need closer logistics or pricing review.



9. Business Story

Olist is a seller-focused retail technology ecosystem, so its business depends on reliable operations across sellers, orders, logistics, payments, and customer experience. The analysis suggests that delivery performance is a major customer-experience signal. Late deliveries are associated with much lower review scores, and delivery risk varies by geography and possibly by seller/category combinations.

This creates a practical business opportunity: instead of only looking at past orders, Olist could monitor operational risk earlier. HVIA could help by building analytics that connect delivery timing, reviews, seller performance, category data, freight burden, and location patterns into one decision system.

10. Proposed HVIA Solution

Logistics & Customer Satisfaction Intelligence Dashboard

The proposed solution is a dashboard and analytics layer that helps Olist monitor operational risk and customer satisfaction. It would start as descriptive analytics and could later become predictive.

- Track late delivery rate by state, seller, category, and month.
- Compare review scores for late vs on-time orders.
- Identify sellers or categories with high order volume but weak satisfaction signals.
- Monitor freight ratio by category and location as a friction indicator.
- Flag risky seller/category/location combinations for deeper review.
- Give management a weekly view of operational bottlenecks and customer experience risks.

11. Outreach Message Draft

Hi Olist team,